

Step 3: k-Sample ANOVA (House Type Effect)

Danish Housing Prices · Mathematical Statistics

Carlos Castillo

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```
library(tidyverse)
dk_clean <- readRDS("../data/dk_clean.rds")

cat("Group sizes:\n")
print(table(dk_clean$house_type))
```

Group sizes:

Apartment	Summerhouse	Townhouse	Villa
16607	9114	9218	48210

1 Statistical Model

Let j index house type ($j = 1, \dots, 4$) and i index observations within each group:

$$Y_{ij} \sim \mathcal{N}(\mu_j, \sigma^2), \quad j = 1, \dots, 4, \quad i = 1, \dots, n_j \quad (\text{independent})$$

The equal-variance assumption σ^2 common across groups must be checked before pooling.

2 Variance Homogeneity — Bartlett's Test

We must perform **Bartlett's test** for $k \geq 3$ groups, applied to the response grouped by factor levels:

$$H_{0\sigma^2} : \sigma_1^2 = \sigma_2^2 = \sigma_3^2 = \sigma_4^2$$

$$Ba = \frac{-2 \ln Q}{C} \sim \chi^2(k-1) \quad \text{under } H_{0\sigma^2}$$

where $C = 1 + \frac{1}{3(k-1)} \left(\sum_{i=1}^k \frac{1}{f_{(i)}} - \frac{1}{f_1} \right)$ is Bartlett's correction factor.

```
cat("Per-group standard deviations:\n")
print(round(tapply(dk_clean$log_price, dk_clean$house_type, sd), 4))

cat("\n--- Bartlett's test ---\n")
bartlett.test(log_price ~ house_type, data = dk_clean)

cat("\nNote: with n ~ 83,000 Bartlett has near-infinite power.\n")
cat("Inspect the SDs directly: if max/min ratio < 2, equal variance\n")
cat("is a reasonable working assumption.\n")
```

Per-group standard deviations:

Apartment	Summerhouse	Townhouse	Villa
0.7425	0.6466	0.7127	0.7574

--- Bartlett's test ---

Bartlett test of homogeneity of variances

data: log_price by house_type

Bartlett's K-squared = 381.8, df = 3, p-value < 2.2e-16

Note: with n ~ 83,000 Bartlett has near-infinite power.

Inspect the SDs directly: if max/min ratio < 2, equal variance
is a reasonable working assumption.

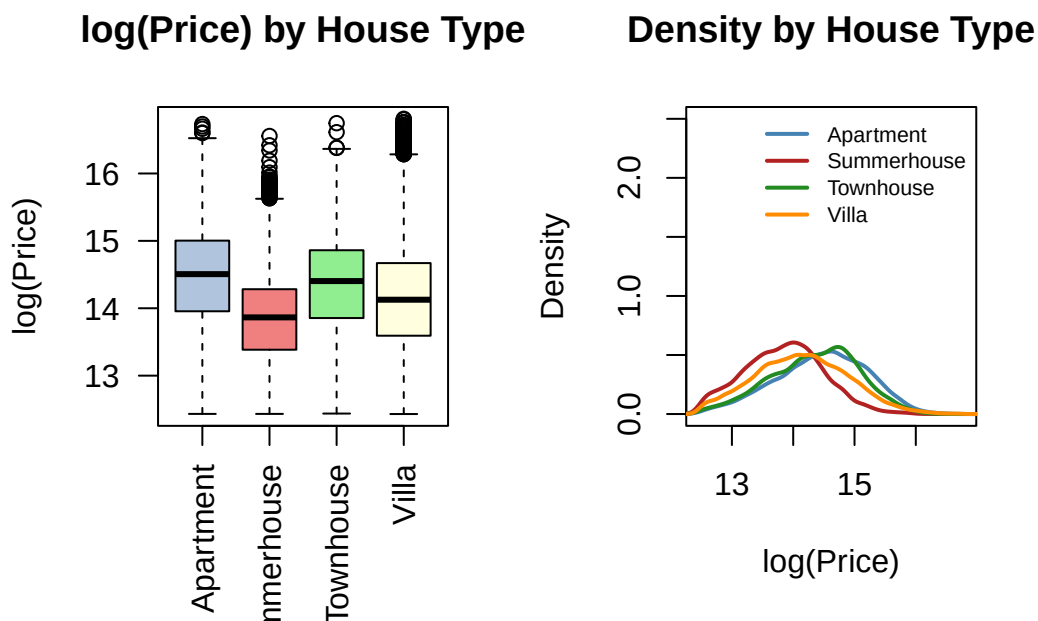
3 Exploratory Plots

```
par(mfrow = c(1, 2))

boxplot(log_price ~ house_type, data = dk_clean,
        col = c("lightsteelblue", "lightcoral", "lightgreen", "lightyellow"),
        main = "log(Price) by House Type",
        xlab = "", ylab = "log(Price)", las = 2)

# Per-group histograms overlaid
types <- levels(dk_clean$house_type)
colors <- c("steelblue", "firebrick", "forestgreen", "darkorange")
xlim <- range(dk_clean$log_price)

plot(NULL, xlim = xlim, ylim = c(0, 2.5),
      main = "Density by House Type",
      xlab = "log(Price)", ylab = "Density")
for (j in seq_along(types)) {
  lines(density(dk_clean$log_price[dk_clean$house_type == types[j]]),
        col = colors[j], lwd = 2)
}
legend("topright", legend = types, col = colors, lwd = 2, bty = "n", cex = 0.7)
```



```
par(mfrow = c(1, 1))
```

4 k-Sample ANOVA — F-Test

4.1 Hypothesis

$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$ vs $H_1 : \text{at least one } \mu_j \text{ differs}$

4.2 F-statistic

$$F = \frac{(\text{RSS}_{M_\infty} - \text{RSS}_{M_1}) / (k - 1)}{\text{RSS}_{M_1} / (n - k)} \sim F(k - 1, n - k) \quad \text{under } H_0$$

```
M_full <- lm(log_price ~ house_type, data = dk_clean) # group-means model
M_null <- lm(log_price ~ 1, data = dk_clean) # common-mean model

cat("--- ANOVA table (full model first per course convention) ---\n")
anova(M_null, M_full)
```

```
--- ANOVA table (full model first per course convention) ---
Analysis of Variance Table
```

```
Model 1: log_price ~ 1
Model 2: log_price ~ house_type
  Res.Df  RSS Df Sum of Sq    F    Pr(>F)
1  83148 47930
2  83145 45302   3    2628.4 1608 < 2.2e-16 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(M_full)
```

Call:

```
lm(formula = log_price ~ house_type, data = dk_clean)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-2.0238 -0.5228  0.0138  0.5150  2.7106
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	14.456061	0.005728	2523.81	<2e-16 ***
house_typeSummerhouse	-0.610331	0.009622	-63.43	<2e-16 ***
house_typeTownhouse	-0.105939	0.009587	-11.05	<2e-16 ***
house_typeVilla	-0.319307	0.006642	-48.08	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7381 on 83145 degrees of freedom

Multiple R-squared: 0.05484, Adjusted R-squared: 0.0548

F-statistic: 1608 on 3 and 83145 DF, p-value: < 2.2e-16

5 Group Estimates

```
cat("Group means on log scale:\n")
grp_means <- tapply(dk_clean$log_price, dk_clean$house_type, mean)
print(round(grp_means, 4))

cat("\nGroup medians on DKK scale (exp of mean log-price):\n")
print(format(round(exp(grp_means)), big.mark = ","))

cat("\nPooled variance estimate s^2:\n")
cat(round(summary(M_full)$sigma^2, 6), "\n")
cat("s =", round(summary(M_full)$sigma, 4), "\n")
```

Group means on log scale:

Apartment	Summerhouse	Townhouse	Villa
14.4561	13.8457	14.3501	14.1368

Group medians on DKK scale (exp of mean log-price):

Apartment	Summerhouse	Townhouse	Villa
"1,897,525"	"1,030,680"	"1,706,785"	"1,378,841"

Pooled variance estimate s^2:

0.544853

s = 0.7381

6 Confidence Intervals for Group Means

```
ci_mat <- confint(M_full)
cat("95% CIs for model parameters (reference = Apartment):\n")
print(round(ci_mat, 4))
```

95% CIs for model parameters (reference = Apartment):

	2.5 %	97.5 %
(Intercept)	14.4448	14.4673
house_typeSummerhouse	-0.6292	-0.5915
house_typeTownhouse	-0.1247	-0.0871
house_typeVilla	-0.3323	-0.3063

7 Interpretation

The F-test compares the **group-means model** M_1 (4 free means) to the **common-mean model** M_∞ (1 mean). Rejection means at least one house type has a different mean log-price — but not which ones.

The `summary()` output shows pairwise contrasts against the reference level (Apartment by default in R). Each t-test there tests $H_0 : \mu_j - \mu_{\text{Apartment}} = 0$.